

CREDIT BUREAU DATASET

Portfolio Project Guide

Data Dictionary • Business Questions • KPIs
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Dataset	credit_bureau_dataset.csv
Records	10,150 (10,000 base + ~150 duplicates)
Columns	22
Period	2022 – 2024
Geography	10 Nigerian States

Column Reference

Column	Type	Description	Example
Customer_ID	String	Unique customer identifier (CB#####)	CB100042
Age	Integer	Customer age in years (18–75)	34
Gender	String	Gender — intentionally inconsistent formatting	Male / M / male
State	String	Nigerian state of residence	Lagos, Rivers
Employment_Status	String	Employment type — intentionally inconsistent	Employed / EMPLOYED
Annual_Income	Float	Annual income in NGN (~3% nulls, outliers)	780,000
Credit_Score	Integer	Credit score 300–850 (~4% nulls)	672
Loan_Type	String	Type of credit facility	Personal, Payday
Loan_Amount	Float	Total loan amount in NGN (contains outliers)	1,500,000
Outstanding_Balance	Float	Current unpaid balance in NGN	945,000
Monthly_Payment	Float	Required monthly payment (~2.5% nulls)	28,500
Number_of_Open_Accounts	Integer	Active credit accounts (1–20)	4
Number_of_Late_Payments	Integer	Lifetime count of missed payments	2
Credit_Utilization_Rate	Float	Used credit / available credit (~3% nulls)	0.4523
Debt_to_Income_Ratio	Float	Annual debt obligations / Annual income	0.3812
Account_Age_Months	Integer	Months since oldest account opened (1–360)	84
Credit_Inquiries_Last_12_Months	Integer	Hard inquiries in last 12 months	3
Default_Status	String	Whether customer defaulted (Yes/No)	No
Delinquency_Status	String	Current payment status	30 Days Late
Risk_Category	String	Derived risk tier (Low / Medium / High)	High
Report_Month	Integer	Month of credit report (1–12)	6
Report_Year	Integer	Year of credit report	2023

Known Data Quality Issues (Intentional)

- **Duplicates:** ~150 duplicate records scattered throughout the dataset.

- **Missing values:** Annual_Income (~3%), Credit_Score (~4%), Employment_Status (~3%), Monthly_Payment (~2.5%), Credit_Utilization_Rate (~3%).
- **Inconsistent formatting:** Gender and Employment_Status contain mixed case, abbreviations, and spacing variants.
- **Outliers:** ~1% of Annual_Income and Loan_Amount records are extreme values (8–20x normal range).

Built-In Business Scenarios

- **Rivers State** carries a ~35% default rate boost — a regional economic shock for geographic analysis.
- **Payday & Personal loans** show significantly higher delinquency rates than other loan types.
- **High-Risk customers** show declining average credit scores across the 2022–2024 reporting period.

SECTION 2 15 BUSINESS QUESTIONS

Questions a Credit Bureau Analyst Could Answer

#	Business Question
1	What is the overall default rate, and how does it vary by state?
2	Which loan types carry the highest delinquency rates?
3	How does credit utilization rate correlate with credit score?
4	What is the average debt-to-income ratio for defaulted vs. non-defaulted customers?
5	Which risk category contributes most to total outstanding balance exposure?
6	Is there a significant relationship between number of late payments and default status?
7	How have credit scores trended year-over-year for High-Risk customers?
8	What is the average loan amount by loan type and employment status?
9	Which age brackets are most likely to be flagged as High Risk?
10	What percentage of customers have a credit utilization rate above 70%?
11	Among customers with 90+ day delinquencies, what is the breakdown by loan type?
12	Is there a correlation between credit inquiries and default probability?
13	How does account age influence risk category assignment?
14	Which state has the healthiest average credit score?
15	What share of total loan exposure sits with already-delinquent customers?

SECTION 3 10 KPIS TO TRACK

Key Performance Indicators

#	KPI	Formula / Definition	Target
1	Portfolio Default Rate	$(\text{Defaulted} / \text{Total}) \times 100$	< 10%
2	Delinquency Rate	$(\text{Not-Current} / \text{Total}) \times 100$	< 15%
3	Average Credit Score	Mean of Credit_Score (excl. nulls)	> 650
4	Average Debt-to-Income Ratio	Mean of Debt_to_Income_Ratio	< 0.40
5	Average Credit Utilization	Mean of Credit_Utilization_Rate	< 0.30
6	High-Risk Customer Share	$(\text{High-Risk Count} / \text{Total}) \times 100$	< 20%
7	Total Outstanding Exposure	SUM(Outstanding_Balance)	Monitor
8	Late Payment Frequency	Mean of Number_of_Late_Payments	< 1.5
9	90+ Day Delinquency Rate	$(90+ \text{ Days Late Count} / \text{Total}) \times 100$	< 5%
10	Credit Score Trend (YoY)	Avg Credit Score this year vs prior year	Positive

Page 1 — Executive Overview

- Card visuals: Total Customers, Portfolio Default Rate, Avg Credit Score, Total Exposure (NGN)
- Stacked bar: Default Status by State — highlights the Rivers State anomaly
- Donut chart: Risk Category distribution (Low / Medium / High)
- Line chart: Avg Credit Score by Report_Year, filterable by Risk_Category

Page 2 — Delinquency & Default Deep Dive

- Matrix: Loan_Type vs Delinquency_Status with conditional formatting
- Bar chart: Default Rate by Loan_Type
- Scatter: Credit_Utilization_Rate vs Credit_Score coloured by Risk_Category
- Gauge: Overall Delinquency Rate vs 15% target

Page 3 — Customer Risk Profiling

- Bar chart: Avg DTI Ratio — Defaulted vs Not Defaulted
- Histogram: Credit Score distribution by Risk_Category
- Table: Top 20 High-Risk customers by Outstanding Balance
- Slicers: State, Loan_Type, Employment_Status, Report_Year

Page 4 — Regional Analysis

- Map visual: Default Rate by State (Nigeria shape map)
- Bar: Avg Credit Score by State
- Bar: Total Outstanding Balance by State
- KPI card: Rivers State Default Rate vs National Average

Suggested DAX Measures

Default Rate (%)

```
DIVIDE(COUNTROWS(FILTER('CreditData', 'CreditData'[Default_Status] = "Yes")),
COUNTROWS('CreditData')) * 100
```

Avg Credit Score

```
AVERAGE('CreditData'[Credit_Score])
```

High Risk %

```
DIVIDE(COUNTROWS(FILTER('CreditData', 'CreditData'[Risk_Category] = "High")),
COUNTROWS('CreditData')) * 100
```

Total Exposure

```
SUM('CreditData'[Outstanding_Balance])
```

Delinquency Rate (%)

```
DIVIDE(COUNTROWS(FILTER('CreditData', 'CreditData'[Delinquency_Status] <> "Current")),
COUNTROWS('CreditData')) * 100
```

BEGINNER

Q1. How many customers are in the dataset?

```
SELECT COUNT(*) AS total_customers FROM credit_bureau;
```

Q2. What is the default rate across the portfolio?

```
SELECT Default_Status, COUNT(*) AS count, ROUND(COUNT(*) * 100.0 / SUM(COUNT(*) OVER ()), 2) AS percentage FROM credit_bureau GROUP BY Default_Status;
```

Q3. How many customers fall into each risk category?

```
SELECT Risk_Category, COUNT(*) AS customer_count FROM credit_bureau GROUP BY Risk_Category ORDER BY customer_count DESC;
```

Q4. What is the average credit score by loan type?

```
SELECT Loan_Type, ROUND(AVG(Credit_Score), 1) AS avg_credit_score FROM credit_bureau WHERE Credit_Score IS NOT NULL GROUP BY Loan_Type ORDER BY avg_credit_score DESC;
```

INTERMEDIATE

Q5. Which states have a default rate above the national average?

```
WITH national_avg AS ( SELECT AVG(CASE WHEN Default_Status = 'Yes' THEN 1.0 ELSE 0 END) AS nat_default_rate FROM credit_bureau ), state_rates AS ( SELECT State, AVG(CASE WHEN Default_Status = 'Yes' THEN 1.0 ELSE 0 END) AS state_default_rate, COUNT(*) AS total_customers FROM credit_bureau GROUP BY State ) SELECT s.State, ROUND(s.state_default_rate * 100, 2) AS default_rate_pct, ROUND(n.nat_default_rate * 100, 2) AS national_avg_pct FROM state_rates s CROSS JOIN national_avg n WHERE s.state_default_rate > n.nat_default_rate ORDER BY s.state_default_rate DESC;
```

Q6. Delinquency breakdown by loan type

```
SELECT Loan_Type, Delinquency_Status, COUNT(*) AS count, ROUND(COUNT(*) * 100.0 / SUM(COUNT(*) OVER (PARTITION BY Loan_Type)), 2) AS pct_within_loan_type FROM credit_bureau GROUP BY Loan_Type, Delinquency_Status ORDER BY Loan_Type, count DESC;
```

Q7. High-utilization, high-DTI customers who haven't yet defaulted

```
SELECT Customer_ID, Credit_Score, Credit_Utilization_Rate, Debt_to_Income_Ratio, Outstanding_Balance FROM credit_bureau WHERE Credit_Utilization_Rate > 0.70 AND Debt_to_Income_Ratio > 0.50 AND Default_Status = 'No' ORDER BY Outstanding_Balance DESC LIMIT 50;
```

Q8. Find duplicate Customer_IDs

```
SELECT Customer_ID, COUNT(*) AS record_count FROM credit_bureau GROUP BY Customer_ID HAVING  
COUNT(*) > 1 ORDER BY record_count DESC;
```

ADVANCED

Q9. Year-over-year credit score trend for High-Risk customers

```
SELECT Report_Year, ROUND(AVG(CASE WHEN Risk_Category = 'High' THEN Credit_Score END), 1) AS
avg_score_high_risk, ROUND(AVG(CASE WHEN Risk_Category = 'Medium' THEN Credit_Score END), 1)
AS avg_score_medium_risk, ROUND(AVG(CASE WHEN Risk_Category = 'Low' THEN Credit_Score END), 1)
AS avg_score_low_risk FROM credit_bureau WHERE Credit_Score IS NOT NULL GROUP BY Report_Year
ORDER BY Report_Year;
```

Q10. Rank states by total outstanding exposure using window functions

```
SELECT State, COUNT(*) AS customers, ROUND(SUM(Outstanding_Balance), 2) AS total_exposure,
ROUND(AVG(Outstanding_Balance), 2) AS avg_exposure, RANK() OVER (ORDER BY
SUM(Outstanding_Balance) DESC) AS exposure_rank FROM credit_bureau GROUP BY State ORDER BY
exposure_rank;
```

Q11. Top 10% most at-risk customers via composite risk score

```
WITH risk_scored AS ( SELECT Customer_ID, Credit_Score, Debt_to_Income_Ratio,
Credit_Utilization_Rate, Number_of_Late_Payments, Outstanding_Balance, (850 -
COALESCE(Credit_Score, 500)) * 0.4 + COALESCE(Debt_to_Income_Ratio, 0) * 100 * 0.3 +
COALESCE(Credit_Utilization_Rate, 0) * 100 * 0.2 + Number_of_Late_Payments * 5 * 0.1 AS
composite_risk_score FROM credit_bureau ), percentiled AS ( SELECT *, NTILE(10) OVER (ORDER BY
composite_risk_score DESC) AS decile FROM risk_scored ) SELECT Customer_ID, Credit_Score,
Debt_to_Income_Ratio, Credit_Utilization_Rate, composite_risk_score FROM percentiled WHERE
decile = 1 ORDER BY composite_risk_score DESC;
```

Q12. Loan type risk profiling — default rate and exposure ranked

```
WITH loan_summary AS ( SELECT Loan_Type, COUNT(*) AS total_customers, ROUND(AVG(Loan_Amount),
2) AS avg_loan_amount, ROUND(AVG(Outstanding_Balance), 2) AS avg_outstanding,
ROUND(AVG(Number_of_Late_Payments), 2) AS avg_late_payments, ROUND(AVG(CASE WHEN
Default_Status = 'Yes' THEN 1.0 ELSE 0 END) * 100, 2) AS default_rate_pct,
ROUND(SUM(Outstanding_Balance), 2) AS total_exposure FROM credit_bureau GROUP BY Loan_Type )
SELECT *, RANK() OVER (ORDER BY default_rate_pct DESC) AS default_rank, RANK() OVER (ORDER BY
total_exposure DESC) AS exposure_rank FROM loan_summary ORDER BY default_rate_pct DESC;
```

Dataset generated synthetically for portfolio use. All customer records are fictional.